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Paleomagnetism and Geochronology of 23 Ma Gabbroic Intrusions in the Keku Strait, Alaska, and Implications for the Alexander Terrane

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Samples of Tertiary gabbro from 24 sites in the Keku Strait, Alaska, help constrain the displacement history of the Alexander terrane. Step heating experiments on a plagioclase separate from these previously undated intrusions indicate a discordant $^{40}Ar/^{39}Ar$ age of 23.1 ± 1.7 Ma. The characteristic magnetization resides in magnetite, is easily isolated by thermal and alternating field demagnetization, and has both normal and reversed polarities. The mean paleomagnetic pole, with no structural correction, is latitude 87.1°N, longitude 141.6°E, A_{95} = 10.1°, with N = 20 sites. This pole indicates insignificant tectonic displacement (0.5° ± 8.2° southward) and rotation (0.6° ± 15.2° counterclockwise). We therefore conclude that any northward displacement or vertical axis rotation of the Alexander terrane, and/or tilting in the vicinity of the Keku Strait must have occurred before 23 Ma.

INTRODUCTION

The collision and accretion of the Alexander terrane to the Stikine terrane profoundly influenced the geologic history of southeastern Alaska and western British Columbia. According to *Monger et al.* [1982] this collision occurred in Late Cretaceous time and an associated orogenic event continued into the early Tertiary. The multitude of Paleocene and mid to Late Cretaceous intrusions of the Coast Plutonic Complex documents this collision [*Monger et al.*, 1982]. However, it is equivocal where the Alexander terrane was located during the collision or its subsequent tectonic history. *Irving et al.* [1985] have suggested that the consistency of shallower than expected paleomagnetic directions from plutons in the Coast Plutonic Complex indicates large-scale northward transport of about 25° of the Alexander, Stikine, and associated terranes in the Intermontane and Insular belts (Figure 1). On the other hand, *Butler et al.* [1989] suggested that these directions, for which there is no control of paleohorizontal, indicate tilting of the Coast Plutonic Complex about 30° to the southwest.

The middle and late Tertiary displacement history of these terranes is less controversial. Eocene plutons intrude the boundary between the Intermontane and Insular belts indicating that differential movement between the terranes had stopped by that time [*Monger et al.*, 1982]. In addition, a number of reliable paleomagnetic studies in western and British Columbia [*Hicken and Irving*, 1977; *Fox and Beck*, 1985; *Bardoux and Irving*, 1989; *Symons and Wellings*, 1989; *Irving and Brandon*, 1990; *Vandall and Palmer*, 1990] and in southern Alaska [*Hillhouse and Gromme*, 1982; *Hillhouse et al.*, 1985; *Coe et al.*, 1985; *Thrupp and Coe*, 1986; *Panuska et al.*, 1990] indicate that the Intermontane and Insular superterranes were in place by about 50 Ma. However, there are virtually no paleomagnetic data from Tertiary rocks on the Alexander terrane in southeastern Alaska. The only existing study was conducted in reconnaissance fashion (eight flows, 29 cores) by *Panuska* [1985] on the ~27 Ma Admiralty Island Volcanics; the results indicated no displacement and no rotation.

In order to elucidate the Tertiary tectonic history of the Alexander terrane, and to compliment paleomagnetic studies of the Late Triassic Hound Island Volcanics [*Hillhouse and Gromme*, 1980; *Haeussler et al.*, this issue], we undertook a paleomagnetic study of Tertiary gabbro in the Keku Strait in southeastern Alaska. We find that displacement, rotation, or tilting in the Keku Strait occurred before 23 Ma and that the gabbro paleomagnetically overprinted a portion of the Hound Island Volcanics on Hound Island and Kuiu Island. We discuss the consequences of intrusion of the Tertiary gabbro upon the paleomagnetism of the Hound Island Volcanics in the accompanying paper [*Haeussler et al.*, this issue].

POINT CAMDEN GABBRO

Introduction

The Keku Strait separates Kuiu from Kupreanof Island in southeastern Alaska (Figure 2). *Muffler* [1967] mapped the geology in the area, finding stratified units that range in age from Silurian to Tertiary, and Tertiary gabbroic dikes and sills. There are large exposures of this Tertiary gabbro at Point Camden on Kuiu Island (Figure 2), and we will informally refer to it as the Point Camden gabbro. The gabbro is olivine and clinopyroxene bearing, and medium grained. *Muffler* [1967] suggested that these intrusions are gently dipping sills up to 450 m thick. He noted that an uncommon magmatic foliation in the gabbro was roughly parallel to the strike of gently dipping bedding in the Tertiary Kootznahoo Formation, which it intrudes, and that the outcrop pattern of the gabbro is roughly parallel to the strike of the Kootznahoo Formation. The large

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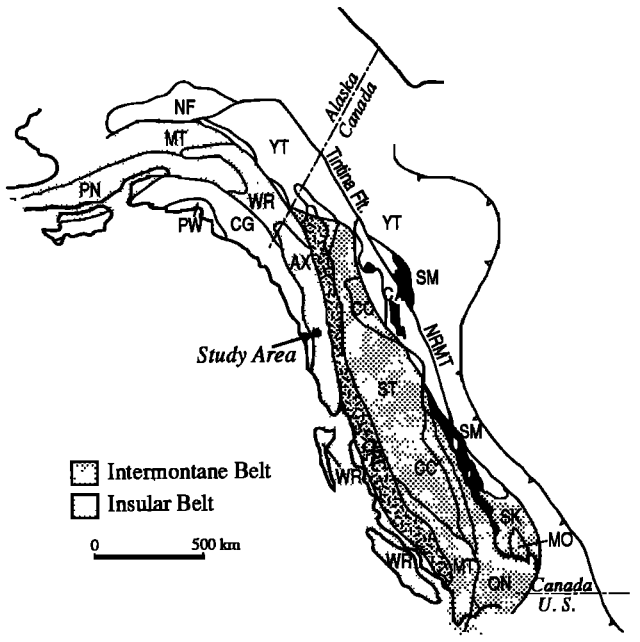


Fig. 1. Major terranes of western Canada and southern Alaska [from *Umhoefer*, 1987]. Terrane abbreviations are AX, Alexander; CA, Cassiar; CC, Cache Creek; CG, Chugach; MT, Miniterranes; MO, Monashee; NF, Nixon Fork; PN, Peninsular; PW, Prince William; QN, Quesnellia; SK, Selkirk; SM, Slide Mountain; ST, Stikine; WR, Wrangellia; and YT, Yukon-Tanana. Slide Mountain is also shown with solid pattern. Coast Plutonic Complex CPC is designated by the hatched pattern. NRMT is northern Rocky Mountain trench. The Insular and Intermontane belts are shown with the broadly spaced and closely spaced dot patterns, respectively.

sills are found only around the southern portion of the Keku Strait, whereas many smaller dikes and sills from 1 to 8 m thick intrude strata on Kuiu Island, which borders the western side of the Keku Strait (Figure 2). On the basis of its inferred age and geographic location the Point Camden gabbro is considered by *Brew and Morrell* [1983] to be the northernmost intrusions of the Kuiu-Etolin volcanic-plutonic belt.

Age and Geochronology

The age of the Point Camden gabbro in the Keku Strait is constrained to be post-Paleocene (work of J. A. Wolfe, discussed by *Brew et al.* [1984]), because it intrudes the Paleocene Kootznahoo formation [*Muffler*, 1967]. K-Ar dates from other more silicic rocks included in *Brew and Morrell*'s [1983] Kuiu-Etolin belt range from 18.5-23.5 Ma with an average of 20.7 Ma [*Douglass et al.*, 1989]. Because the Point Camden gabbro is more mafic than most rocks in the Kuiu-Etolin belt, and for the greatest tectonic significance of our paleomagnetic results we decided to obtain an isotopic age for the gabbro. An accurate date would also allow us to to constrain the thermal history and the age of overprints we found in the Hound Island Volcanics [see *Haeussler et al.*, this issue].

Thin section examination of a hand sample from paleomagnetic site 76 indicated fresh plagioclase, suitable for $^{40}Ar/^{39}Ar$ dating. From this sample, we obtained a plagioclase separate through conventional crushing, sieving and magnetic separation methods. For $^{40}Ar/^{39}Ar$ analysis, 1.0770 g of plagioclase from the 254 to 423 μm size fraction was incrementally degassed at Princeton University using the facilities and procedures outlined by *Onstott and Peackock* [1987]. Summary data are given in Table 1. The interlaboratory standard Mmhb-1 (520.4 Ma [*Samson and Alexander*, 1987]) was used as a fast neutron fluence monitor ($J = 5.10339 \times 10^{-3} \pm 1.83314 \times 10^{-4}$).

Incremental heating in 15 steps between 494° and 1200°C produced a discordant apparent age spectrum (Figure 3), with initially high apparent ages in the first two steps (below 600°C). Subsequently, all steps define a relatively flat spectrum, with only the 848°C step departing from a plateau (e.g., according to the criteria of *Lanphere and Dalrymple* [1978]) in the sense that their apparent ages are within 2σ of the mean's 2σ envelope. Within this segment, 37% of the ^{39}Ar was released in the first step, at 648°C, yielding a date of 22.2 ± 1.6 Ma. Subsequent steps contained much less Ar, and the uncertainties are much larger as a result. A "plateau" date, calculated as the weighted (by inverse variance) mean of dates from all 13 plateau steps, is 23.1 ± 1.7 Ma (1σ) . Uncertainty in the plateau date includes the contribution from uncertainty in the J value. The plateau date, and its uncertainty, is dominated by the relatively large initial step with its relatively small apparent age uncertainty.

The large (5.9%) uncertainty in the plateau date is primarily a result of the low K concentration in the plagioclase but is exacerbated by the high Ca content, which contributes to error through uncertainty in the proportion of nucleogenic (versus atmospheric) ^{36}Ar. The integrated $^{37}Ar_{Ca}/^{39}Ar_K$ is 23.15 ± 0.08, corresponding to a bulk Ca/K of 42.13.

The anomalously high apparent ages for the two lowest temperature steps are reminiscent of patterns produced by excess ^{40}Ar located in nonretentive crystal chemical sites. To determine whether any such excess ^{40}Ar is discernable in the plateau steps, the data were plotted on a $^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$ isotope correlation diagram. Ten of the 13 "plateau" steps define an array that yields an isochron intercept date of 21.9 ± 1.3 Ma (1σ) with an initial $^{40}Ar/^{36}Ar$ intercept of 300.3 ± 6.3, mean squared weighted deviation (MSWD) = 0.88. The initial $^{40}Ar/^{36}Ar$ isotopic composition is not significantly different from that of air (295.5), and (consequently) the isochron intercept date is indistinguishable from the plateau date.

The 23.1 ± 1.7 Ma "plateau" date is a reasonable estimate of the age of intrusion at site 76 because of the rapid cooling rate inferred for the gabbro (discussed below). We infer that intrusion of all of the Point Camden gabbro occurred around 23 Ma. The 23-Ma isotopic age from the gabbro is consistent with *Brew and Morrell's* [1983] contention that the Point Camden gabbro is part of the Kuiu-Etolin belt. *Muffler* [1967] believed that the Point Camden gabbro is genetically related not only to basalts that overlie the Kootznahoo Formation, which *Brew and Morrell* [1983] considered part of the Kuiu-Etolin belt, but also to the Admiralty Island Volcanics, which lie 30 km north of Hound Island and have a whole rock K/Ar age of 27 Ma [*Brew*, 1988]. Therefore, the 23-Ma isotopic age we obtain is consistent with all geologic arguments for the age of the Point Camden gabbro.

Structural Geology

The structural geology of the Keku Strait is not well understood despite detailed mapping, because many important contacts and faults are submerged (Figure 2) [*Muffler*, 1967]. The major deformational episode in the study area probably

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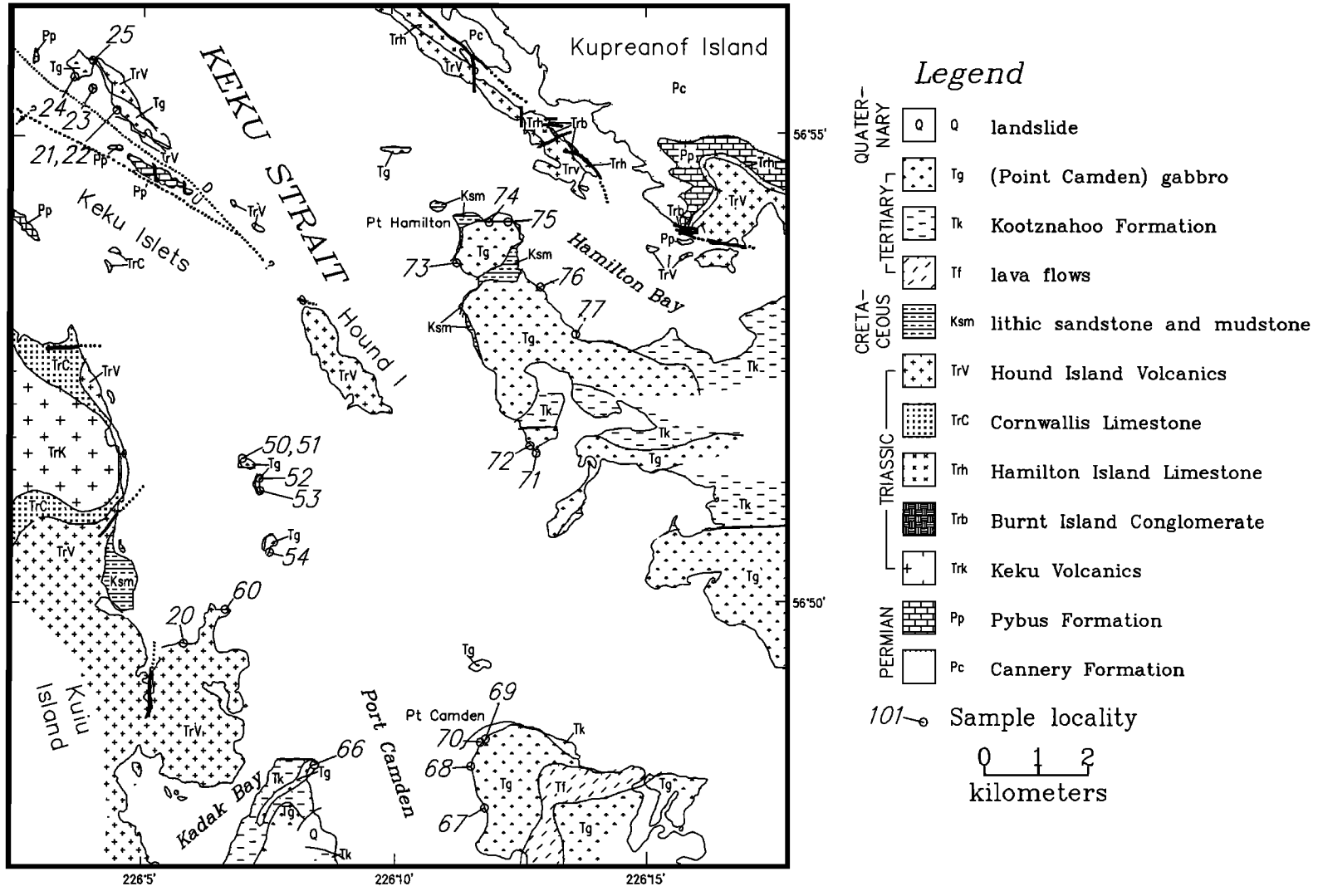


Fig. 2. Geologic map of the Keku Strait showing the location of paleomagnetic sites [from *Muffler*, 1967].

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TABLE 1. $^{40}Ar/^{39}Ar$ Analytical Data From the Point Camden Gabbro Site 76

T, °C	Cum ^{39}Ar	$^{37}Ar_{Ca}/^{39}Ar_K$	$^{40}Ar^*/^{39}Ar_K$	Date ± 1σ, Ma
494	0.1378	15.2669	30.75400	263.0 ± 20.4
599	0.3349	32.1859	6.74763	61.1 ± 3.6
64?	0.5825	25.4650	2.42695	22.2 ± 1.6
654	0.6122	19.5877	2.30698	21.1 ± 31.1
701	0.6435	22.8108	7.20731	65.2 ± 25.2
748	0.6961	24.7158	2.90469	26.5 ± 13.9
800	0.7323	30.7388	4.22313	38.5 ± 34.9
848	0.7807	27.3705	5.38439	48.9 ± 9.5
898	0.8121	14.2268	1.40172	12.9 ± 12.2
949	0.8561	13.1325	1.94005	17.8 ± 12.3
949	0.8798	9.2483	1.89230	17.3 ± 8.0
1000	0.9021	11.9096	3.20838	29.3 ± 6.4
1055	0.9214	12.0731	2.23181	20.4 ± 4.8
1101	0.9440	13.9812	2.41614	22.1 ± 7.2
1200	1.0000	23.3080	3.51233	32.1 ± 7.2

Cum ^{39}Ar is the cumulative fraction of ^{39}Ar released. Subscripts denote parent element for nucleogenic Ar; $^{40}Ar^*$ is radiogenic ^{40}Ar. Uncertainties given for individual steps' dates do not include contribution from uncertainty in J value.

occurred in mid-Cretaceous time (probably between ~100 and 90 Ma, see *Haeussler et al.* [this issue] for arguments). However, some deformation is probably Late Cretaceous age or younger. The Keku Islets form several prominent northwest-to-southeast trending linear strings of islands, which suggests a structural control on their orientation. In addition, the bathymetry in the Keku Strait has a northwest-southeast trend, and there are many northwest-southeast trending lineaments elsewhere in the geomorphology of southeastern Alaska. It is likely that many of these lineaments are post-90 Ma right-lateral strike-slip faults, which is suggested by outcrop-scale faults with the same orientation and a dextral sense of offset [*Haeussler*, 1991]. In the Keku Strait, the similarity of adjacent units precludes any large amount of strike-slip movement on the faults. Also, the inferred faults all indicate at least some vertical movement, but without a consistent direction of offset [*Muffler*, 1967]. *Muffler* [1967] concluded that there has not been major transport on any of the faults in the Keku Strait.

Deformation in the Keku Strait younger than Paleocene in age is suggested by southward dip of the Kootznahoo Formation. Bedding attitudes of the Kootznahoo Formation from *Muffler*'s [1967] map have a mean strike of N86°E and dip of 13°S (Figure 4). Therefore the Point Camden gabbro may have been tilted to the south, if it was intruded before these rocks were tilted. Our paleomagnetic evidence, discussed later, argues against this possibility.

The only faults in the vicinity of the sampling localities are the two shown in the northwestern corner of Figure 2 [*Muffler*, 1967]. The northeastern fault is inferred from the juxtaposition of Permian and Late Triassic rocks, and some northeast-side-down movement is required. The other fault is inferred from the linear trend of the islands and cannot have significant horizontal or vertical offset.

The constraints on the timing of movement on these faults is poor. The Admiralty Island Volcanics, which lie 30 km to the north on southern Admiralty Island, are approximately coeval with the Point Camden gabbro and are not cut by any faults with a northwest-southeast trend, like those in the Keku Strait. This relationship suggests that movement on the NW-SE striking faults occurred prior to extrusion of the Admiralty Island Volcanics 27 Ma. We know of no other constraints on the timing of Tertiary faulting in the Keku Strait.

PALEOMAGNETISM

Sample Collection and Measurement

We collected 144 samples from 24 sites (six samples/site) in the Point Camden gabbro (Figure 2). We used a portable coring device to cut the samples, which were oriented using a magnetic compass and inclinometer. Magnetic compass readings were checked with the solar compass and by back siting. In the laboratory, we performed detailed thermal demagnetization in air, with a minimum of nine and an average of 12 demagnetization steps on each sample. All measurements were made using a 2G three axis cryogenic magnetometer in a shielded room at the University of California, Santa Cruz. We chose to use thermal demagnetization on the samples, because the characteristic component of magnetization was somewhat better defined than alternating field (AF) demagnetization made on companion samples from the same core (compare Figures 5*a* and 5*b*). In addition, AF demagnetization of some coarse-grained igneous rocks can impart a spurious component of magnetization, making interpretation of the demagnetization data more difficult.

Magnetic Mineralogy

All lines of evidence are consistent with Ti-poor titanomagnetite as the carrier of remanence in the Point Camden

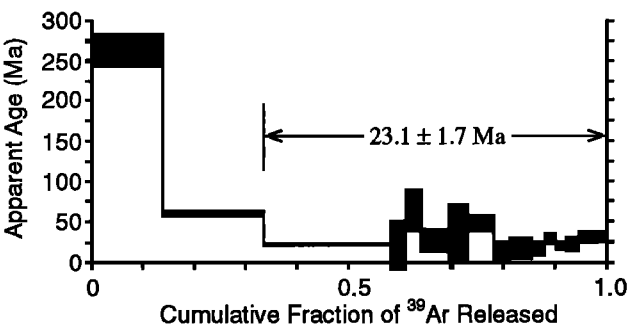


Fig. 3. Apparent age spectrum for plagioclase separate from site 76, showing steps that were included in the plateau date. Error bars are ± 1 standard deviation, and do not include the uncertainty in the *J* value.

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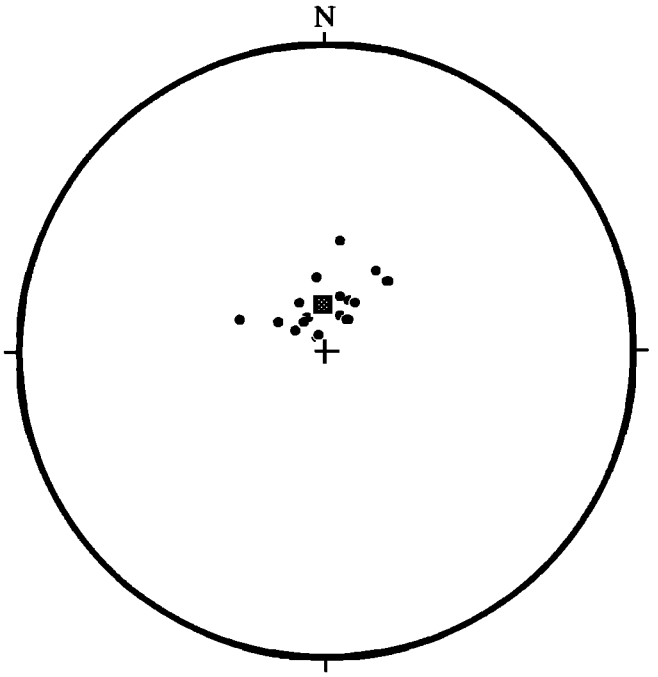


Fig. 4. Stereographic projection of poles to bedding of the Kootznahoo Formation in the Keku Strait. The mean orientation is strike N86°E, dip 13°S, and the pole to this plane is shown with a box.

gabbro. First, laboratory unblocking temperatures for all samples are less than 600°C (see examples in Figure 5), which indicates the carrier of the remanence is magnetite. Observed blocking temperatures between 580°C (the Curie temperature of pure magnetite) and 600°C are almost certainly an artifact of longitudinal temperature variations of up to 30°C in our Schoenstedt model TSD-1 oven. Plots of the intensity of remanence versus demagnetization temperature often show a very large drop in intensity in the range of 400°-520°C (Figure 5), which suggests the magnetic carrier is titanomagnetite rather than pure magnetite. Alternatively, this effect could also be due to a range of magnetic grain sizes with different blocking temperatures. Second, coercivities of less than 100 mT for samples that were AF demagnetized (see example in Figure 5*b*) are also consistent with magnetite or titanomagnetite as the remanence carrier. Third, IRM isothermal remanent magnetization (IRM) acquisition curves (Figure 6) suggest the magnetic carrier is magnetite or titanomagnetite. All samples show a rapid gain in intensity between 0 and 150 mT applied field and reach a plateau at 250 mT, indicating that Ti-poor titanomagnetite is the only magnetic phase. We find the intensity of the IRM is about the same in some heated and unheated sample pairs, which indicates that no alteration of the magnetic mineralogy occurred during heating (samples 86H0169, 87H0342, and 87H0465 in Figure 6). In other cases we find the heated samples have a lower saturation remanence than the unheated samples (samples 87H0371 and 87H0485 in Figure 6). Finally, magnetite is the only magnetic mineral we observe in polished thin sections of samples. Generally, the visible magnetite grains are equant, euhedral to subhedral, and of the "composite" type [*Haggerty*, 1976]. Coarse ilmenite exsolution lamellae are faintly seen in less than half of the studied samples and may indicate rapid cooling. There is no evidence of alteration of any of the opaque minerals or for migration of titanium into new mineral phases such as sphene. We conclude that Ti-poor titanomagnetite is the most likely carrier of remanence in the Point Camden gabbro.

Analysis of Directions

The demagnetization path defined on vector diagrams revealed one or two components (Figures 5 and 7). The low-temperature component commonly has scattered directions. Low unblocking temperatures and low coercivities are consistent with a viscous component acquired during storage; however, blocking temperatures of up to 350°C in some samples seem too high for a viscous remanent magnetization (VRM) acquired at room temperature for less than 1.5 years. It is also unlikely that this scatter is due to lightning strikes, because these rocks were collected at sea level and electrical storms are rare in southeastern Alaska. Less commonly, this direction is parallel to the present day geomagnetic field (PDF) direction, but it can be difficult to distinguish an early Miocene normal-polarity direction from the PDF direction.

The high-temperature componènt is usually well defined and is oriented north and steeply down or south and moderately to

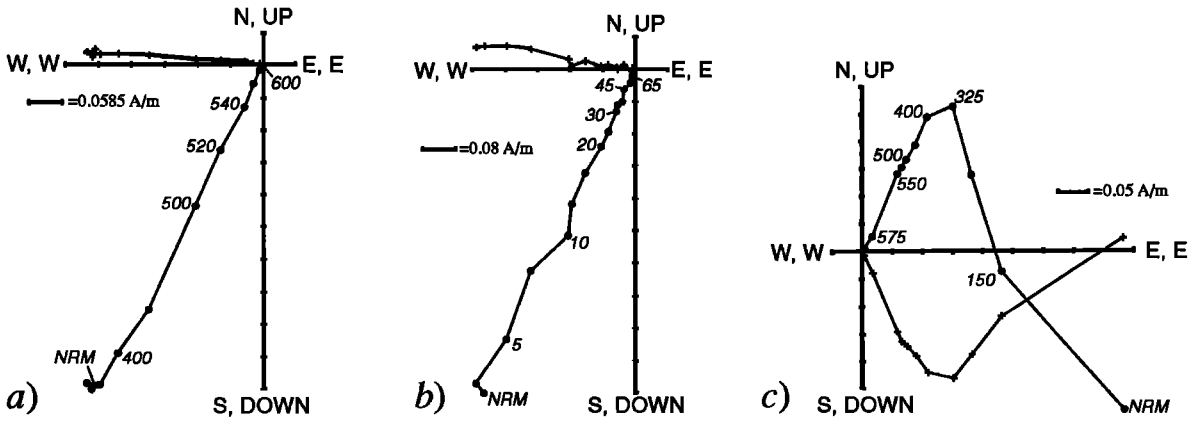


Fig. 5. Zijderveld diagrams showing typical demagnetization behavior of samples of the Point Camden gabbro. Numbers next to a data point give either the temperature (in degrees Celsius) or the applied alternating field (AF) field (in milliteslas). (*a*) Normal polarity, thermally demagnetized sample. Note the relatively narrow range in unblocking temperatures. Sample 86H0225A is from site 20. (*b*) AF demagnetized companion sample to that shown in Figure 5*a*. Sample 86H0225B is from site 20. (*c*) Reversed polarity. Sample 87H0445A is from site 88.

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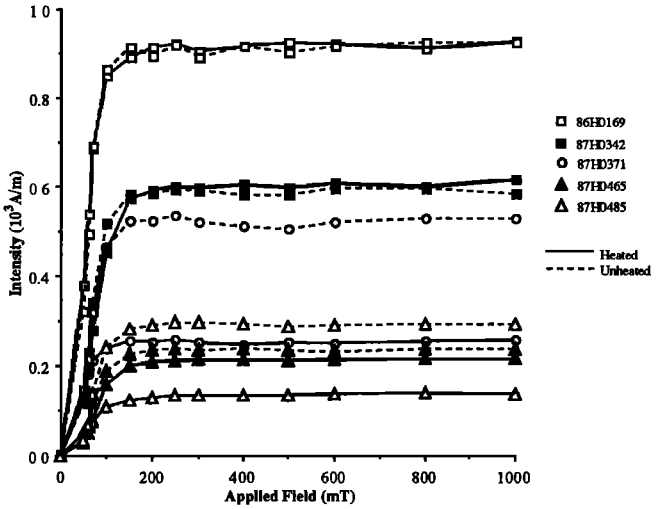


Fig. 6. Isothermal remenant magnetization acquisition curves of pairs of typical heated and unheated samples. The curves for both samples were normalized to the one with the highest value. Each heated and unheated pair of samples is shown with an identical symbol.

vertically upward (Figures 5 and 7). For individual samples, we calculated the orientation of the high-temperature component by a least squares fit of the portion of the vector diagram that decayed univectorially toward the origin of the plot. Between three and 10 points, with an average of 6.4 points, were used to define this straight-line segment. Within an individual site, the best fit directions cluster tightly. The Fisher precision parameter k is over 100 for all but one site, with values as high as 3400. The range in laboratory unblocking temperatures within a site is commonly small (often <40°-50°C). If the rocks cool through this temperature range in a few thousand years or less, then a single specimen or locality would not completely average secular variation. For example, *Dodson et al.* [1978] found that samples from vertical cores through the Tatoosh intrusion, in the Washington Cascade Range, display a paleomagnetic record similar to present secular variation records, and record a polarity transition of the Earth's magnetic field. *Dodson et al.* [1978] contended that because such a record is present, cooling rates in the pluton were fast enough to go through the blocking temperature range of the samples in a short period of time. Simple conductive cooling models for the Tatoosh are precluded, because thermal modeling indicates it would take 80,000 years for samples 500 m from the margin of the pluton to cool from 580°C to 560°C; thus cooling by fluids seems to be called upon.

Taylor [1971] has shown that the oxygen isotope ratios for several intrusions in the Cascades indicate contamination by meteoric water. Therefore some intrusions are at least partially cooled by circulating groundwater. *Taylor* [1971] suggested that a hydrothermal convection system begins as soon as the pluton is intruded, but the groundwater cannot aid cooling of the body until there is a solidified margin in which cracks can open and allow circulation. Thus *Dodson et al.* [1978] developed conductive cooling models that take into account crack propagation through a cooling intrusion with a hydrothermal convection system. This model may also explain the paleomagnetic directions in the Point Camden gabbro. *Muffler* [1967] estimated the sills(?) of the Point Camden gabbro to be about 500 m thick. Even if the gabbroic intrusions were twice as thick, it would take no longer than 1000 years to cool from 580° to 500°C in the case of conductive cooling to a convective surface [*Dodson et al.*, 1978]. The actual range in blocking temperatures within our samples is usually less (commonly 40°-50°C), and the intrusion is probably thinner, and thus the samples at a locality may cool through their blocking temperatures in much less than 500 years. This evidence indicates that small epizonal intrusions, such as the Point Camden gabbro, can cool fast enough to record secular variation of the Earth's magnetic field, but only if the remanence is a thermal remanent magnetization (TRM).

Additional evidence for this hypothesis comes from examining the directions in sites 67 through 70. These four sites lie along the coastline to the south and west of Point Camden on Kuiu Island (Figure 2) and have directions suggestive of an excursion; the directions are shallower at sites further to the north (see Figure 7 and Table 2). S. H. Lee and M. W. McElhinny (work discussed by *Merrill and McElhinny* [1983]) considered virtual geomagnetic poles (VGPs) that are greater than 40° from the mean to be transitional directions. Site 70 has such a VGP. We also note that the range in blocking temperatures for this site is less than 20°C, which suggests to us the rocks cooled through this temperature range in a short amount of time (less than a few thousand years), allowing the rocks to record the transitional direction. At site 69, furthest to the north, the direction is again close to the mean. These data suggest to us that a cooling front was moving through this portion of the intrusion during an excursion or reversal of the geomagnetic field as the rocks at sites 67 through 70 were cooling through their blocking temperatures.

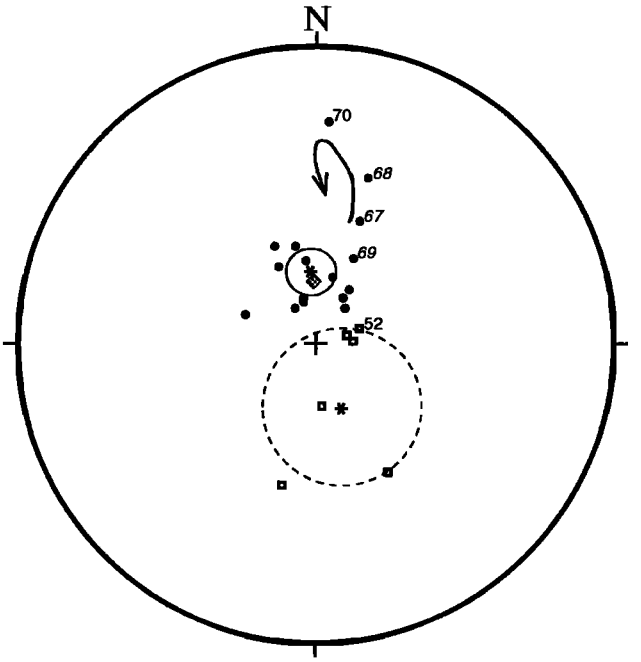


Fig. 7. Stereographic projection of site mean directions of the Point Camden gabbro. Lower hemisphere directions are dots, and upper hemisphere directions are open squares. The mean of all directions, with reversed directions inverted to the lower hemisphere, is shown with an asterisk and its associated 95% circle of confidence. Mean directions from sites 52 and 70, labeled, are not included in calculation of the mean, because we believe they record transitional directions (see text). The mean directions from sites 67, 68, 70, and 69 may define an excursion, as indicated with the arrow (see text). The mean of the upper hemisphere directions is shown with an asterisk and its dashed 95% circle of confidence. The grey diamond shows the expected 20-Ma direction for North America from *Hagstrum et al.* [1987].

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TABLE 2. Paleomagnetic Data From the Point Camden Gabbro.

Site	N	P	R	α_{95}	k	I	D
20	6/6	N	5.97891	4.4	237.1	69.2	293.0
21	6/6	N	5.98325	3.9	298.6	74.5	26.6
22	6/6	N	5.97145	5.1	175.2	71.3	34.4
21&22	12/12	N	11.94749	3.0	209.5	72.9	30.9
23	6/6	N	5.98851	3.2	435.3	77.6	39.3
24	6/6	N	5.99333	2.4	749.2	75.5	31.4
25	6/6	N	5.99697	1.6	1652.4	78.5	343.8
50	5/6	R	4.98687	4.4	304.6	-76.8	80.1
51	6/6	R	5.96503	5.6	143.0	-81.9	96.3
50&51	11/12	R	10.93767	3.6	160.4	-79.7	87.0
52*	6/6	R	5.97228	5.0	180.4	-77.6	70.2
53	6/6	R	5.96706	5.5	151.8	-81.2	74.7
54	6/6	R	5.98525	3.6	338.6	-50.0	193.5
60	6/6	N	5.96592	5.5	146.7	67.4	353.1
66	6/6	R	5.83528	12.4	30.4	-49.1	150.3
67	6/6	N	5.99520	2.1	1041.7	54.3	19.7
68	6/6	N	5.97486	4.8	198.9	41.6	17.6
69	6/6	N	5.98823	3.3	424.8	71.2	14.2
70*	6/6	N	5.99206	2.7	629.5	26.2	3.0
71	6/6	N	5.99458	2.0	1107.3	78.6	329.0
72	6/6	N	5.99853	1.1	3405.9	76.9	345.5
73	5/6	R	4.98761	4.3	322.8	-72.7	173.5
74	4/6	N	3.99479	3.8	575.9	61.1	337.2
75	6/6	N	5.99176	2.7	607.1	66.6	334.5
76	5/6	N	4.99865	1.4	2961.6	62.8	348.5
77	5/6	N	4.99772	1.8	1756.7	64.6	23.7
Mean	20	N	19.24428	6.6	25.1	70.5	355.7

N is the number of stable samples in a locality out of the total number collected; P, polarity, with N, normal polarity, and R, reversed polarity; *I*, in situ declination; *D*, in situ declination. The average site latitude and longitude is 56.9°N, 226.0°E.

*These sites were rejected because their VGP's lie > 40° from the mean paleomagnetic pole.

If these subhorizontal intrusions are sill-like, as proposed by *Muffler* [1967], an excursion that is recorded laterally within the sill indicates that cooling of the sill did not occur along subhorizontal isotherms. The Point Camden gabbro is not laterally extensive, and therefore this result may be due to lower temperatures near the distal margins of the sills.

We drilled two pairs of sites close to each other in order to investigate the variation in remanence directions of nearby sites. Sites 21 and 22 are located 10 m apart, and sites 50 and 51 are located 25 m apart. In both cases, both sites at a locality have very similar directions; they are not different at the 95% confidence level. These data imply that each pair of sites were cooled through their blocking temperatures at about the same time. When we calculated site mean directions, we averaged sites 21 and 22 and 50 and 51 (see Table 2). At adjacent sites further apart we do not find similar directions. Sites 69 and 70, discussed above, are about 200 m apart, yet show significantly different directions at the 99.9% confidence level.

Discussion of Remanent Magnetism

For several reasons, we conclude that the high-temperature component in the Point Camden gabbro is primary. First, the two characteristic components of magnetization are easily isolated, and there is no evidence for unresolved components of magnetization. Second, we find both polarities, whereas overprinted rocks most often record only one polarity. Third, as discussed previously, we believe that sites 67-70 are most easily interpreted as recording transitional directions of the Earth's magnetic field. We consider it unlikely that paleomagnetically overprinted rocks would have directions like these, although if the high-temperature component was not well isolated (which does not appear to be the situation), or if an overprint had overlapping blocking temperatures with the characteristic directions, similar directions could be produced. Fourth, the magnetic mineralogy we observe in thin section appears pristine and unaltered. Finally, there are no known widespread thermal or tectonic events younger than 23 Ma in southeastern Alaska, and thus there is no reason to suspect that the Point Camden gabbro should be overprinted. There are no metamorphic rocks younger than the Point Camden gabbro in southeastern Alaska, and the only younger igneous rocks are flat-lying lava flows spatially restricted to southeastern Kupreanof Island. For these reasons we conclude the high-temperature components of magnetization in the Point Camden gabbro are primary.

We consider all site mean directions that we have obtained to be reliable, but we believe that transitional directions contribute noise to the data set and should not be used for determining a paleomagnetic pole. Using the criterion of S. H. Lee and M. W. McElhinny (work discussed by *Merrill and McElhinny* [1983], mentioned earlier, we find two transitional site mean directions: sites 52 and 70. We do not use these in our final calculations, although the conclusions of our study are not altered whether or not they are included.

As discussed above, we believe that individual sites in the Point Camden gabbro do not average secular variation, and thus we need to evaluate if the collective site mean VGPs average secular variation. The expected value of the angular standard deviation *S*, a measure of the dispersion of the VGPs, at the paleolatitude determined for this site during the last 5 m.y. is 18.5, and for the interval from 22.5 to 45 Ma it is 22 [*McFadden et al.*, 1991]. The *S* value we determine with the VGPs from the Point Camden gabbro is 24, which is close to

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(but slightly more scattered than) the expected value. The work of *McFadden et al.* [1991] shows a greater amount of VGP scatter for the interval from 22.5 to 45 Ma than for the past 5 m.y. The S value we determine is within 9% of the expected value for the interval from 22.5 to 45 Ma. In addition, the data pass a reversal test. The means of the normal and reversed polarity directions are not significantly different at 95% confidence. This reversal test is positive whether or not we reject any sites that may be transitional directions. It is noticeable that the reversed-polarity site mean directions are more scattered than the normal-polarity site mean directions, which could suggest that the reversed-polarity directions are not well isolated. However, we emphasize that vector diagram plots of samples with reversed directions show that the characteristic direction is clearly defined. The high value of the Fisher precision parameter k for all but one of these sites (Table 2) also supports the assertion that we have isolated the primary reversed polarity direction. Alternatively, if a normal polarity overprint had overlapping blocking temperatures with a reversed polarity component, the scatter might be increased. In conclusion, we consider it likely that our study provides a good estimate of the mean direction.

TECTONIC IMPLICATIONS

We obtained a mean paleomagnetic pole, determined from the mean of the site-mean VGPs with no structural correction, of latitude = 87.1°N, longitude = 141.6° E, A_{95} = 10.1°, with N = 20 sites. For this average, we have used all the site mean directions, except for sites 52 and 70, and we combined sites 21 and 22, and 50 and 51 together. As discussed earlier, portions of the Point Camden gabbro appear to be large sills intruded subparallel to bedding of the Kootznahoo Formation. Therefore a "bedding" correction for the Point Camden gabbro may be provided by the orientation of the Kootznahoo Formation. The bedding attitudes of the Kootznahoo Formation from *Muffler*'s [1967] map that have a Fisher mean strike of N86°E and dip of 13°S (Figure 4). The "tilt-corrected" mean paleomagnetic pole is latitude = 66.7°N, longitude = 222.2°E, A_{95} = 10.1°.

In order to assess the tectonic significance of these paleomagnetic poles, we compare them to the early Miocene North American paleomagnetic pole at approximately 20 Ma determined by *Hagstrum et al.* [1987]. They combined the 20 Ma pole of *Irving and Irving* [1982] with two poles from paleomagnetic studies in the Sierra Madre Occidental, for a mean pole of latitude = 87.4°N, longitude = 129.7°E, A_{95} = 3.0°. Assuming that the present horizontal is the same as the paleohorizontal, we calculate an insignificant southward displacement of 0.5° ± 8.2° and counterclockwise rotation of 0.6° ± 15.2°. Retaining the site mean directions that are excluded as transitional has essentially no effect: the displacement is 4.4° ± 8.7° southward and the rotation 4.1° ± 16.1° counterclockwise. If the paleomagnetic pole is "tilt corrected" using the orientation of the Kootznahoo Formation, a southward displacement of 23.5° ± 8.2° and counterclockwise rotation of 4.0° ± 52.6° are inferred. We calculated the uncertainty of the displacement and rotation using the methods of *Demarest* [1983] using the factor C = 0.78 consistently.

The simplest interpretation of this result is that the intrusions have not been significantly tilted and the Alexander terrane has remained in its present position and orientation with respect to North America since cooling 23 m.y. ago. This result is the first thorough paleomagnetic study in southeastern Alaska on late Oligocene/early Miocene rocks; the result is not surprising and agrees with other paleomagnetic and geologic constraints. First, the only major displacement recognized on faults younger than 23 Ma in southeastern Alaska is on the Queen Charlotte-Fairweather fault system and on the Chatham Strait fault, both of which lie west of the Keku Strait and therefore would not affect its paleolatitude. In addition, the Tintina-Rocky Mountain Trench fault system, which lies to the east in British Columbia (Figure 1), does not have significant Neogene offsets [*Gabrielse*, 1985], consistent with our paleomagnetic data. Second, plutons that intrude both the Alexander and Stikine terranes are older than 23 Ma, indicating that movement between the terranes had stopped by that time. Third, as outlined in the introduction, a number of reliable paleomagnetic studies in western British Columbia and in southern Alaska indicate that the Intermontane and Insular superterranes were in place by about 50 Ma. Finally, the previously mentioned reconnaissance paleomagnetic study (eight flows, 29 cores) on the coeval Admiralty Island Volcanics, which outcrop 30 km to the north of the Point Camden gabbro, also indicates no displacement and no rotation [*Panuska*, 1985].

The alternative interpretation, that the Alexander terrane has moved southward 23.5° in the last 23 m.y., is untenable with any paleomagnetic, structural, or plate motion data, and thus we reject it. A corollary interpretation is that the Kootznahoo Formation was tilted between Paleocene and Miocene time, which is a significant constraint on the Tertiary structural history of southeastern Alaska.

An alternative allowable interpretation is that bedding of the Kootznahoo Formation had an initial dip. The Kootznahoo Formation is an alluvial fan deposit with medium to very coarse grained sandstones and common conglomerate clasts and lenses [*Loney*, 1964]. *Loney* [1964, p. 93] describes the deposition of the Kootznahoo Formation as follows: "The rugged mountains cut in the folded Mesozoic and Paleozoic rocks adjacent to this basin poured down coarse detritus..., probably forming steep alluvial fans through which cut steep gradient streams." An initial dip of 13° is on the high side for an alluvial fan [*Kochel and Johnson*, 1984] but is not unheard of. Thus we believe that some initial dip is possible and perhaps probable. In addition, it is notable that the Kootznahoo Formation dips to the south, in contrast to the major structural grain of southeastern Alaska, which trends NNW - SSE. This suggests that either there was a change in the general tectonic regime between the time the Kootznahoo Formation was deposited and when it was tilted, or possibly some of the dip of the Kootznahoo Formation is primary.

A final implication of the paleomagnetic results from the Point Camden gabbro is if the Coast Plutonic Complex was tilted as *Butler et al.* [1989] propose, the tilting did not affect the Keku Strait. Either tilting did not occur along the portion of the Coast Plutonic Complex in southeastern Alaska, or tilting occurred before the Point Camden gabbro was intruded, or uplift of the Coast Plutonic Complex only affected the region to the northeast of the Keku Strait. Further studies of mid to late Tertiary intrusions from within and near the Coast Plutonic Complex will add to our understanding of how these rocks were uplifted and what the regional consequences of uplift are.

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